Applications of Integration

Dr. Vindya Bhat

MATH 1851

Tutorial 6

香 港 大 學
THE UNIVERSITY OF HONG KONG

Tutorial Objectives

- Express volume of a solid of revolution as an integral
- Use disk, washer and cylindrical shells methods to calculate volume
- Calculate arc length and surface area as an integral

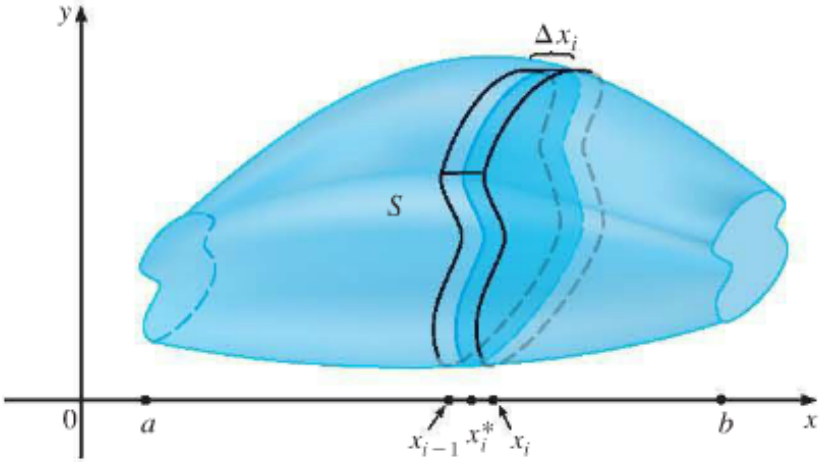
Tutorial Timeline

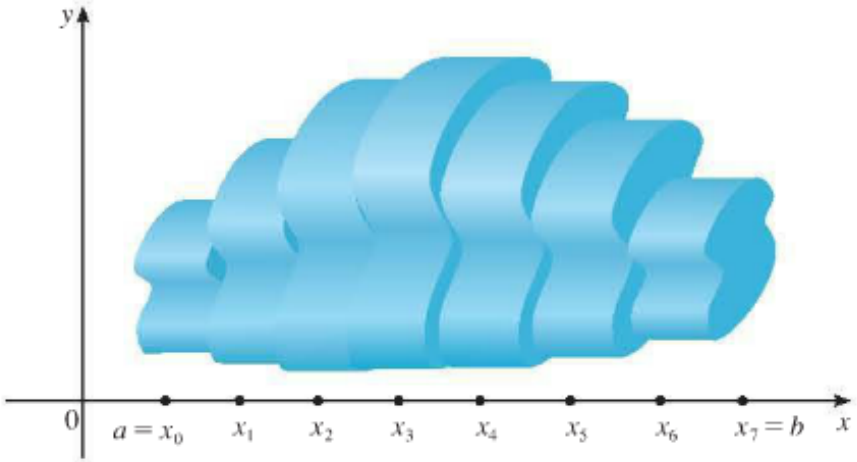
Topic	Time
Volumes	30 minutes
Arc Length	5 minutes
Surface Area	5 minutes
Return Exams	10 minutes

Definition

Let S be a solid that lies between $x = a$ and $x = b$. If the cross-sectional area of S in the plane P_x, through x and perpendicular to the x-axis, is $A(x)$, where A is an integrable function, then the **volume** of S is

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} A(x_i^*)\Delta x = \int_a^b A(x)\,dx.$$



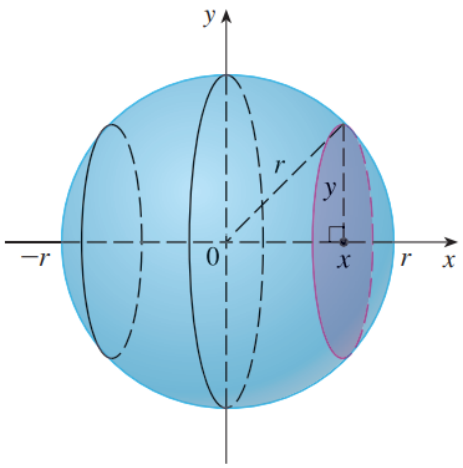


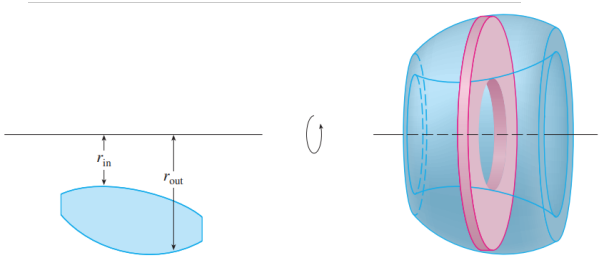
Disk/Washer Method

Definition

Solids obtained by revolving a region about a line is called a **solid of revolution**.

- $V = \int_a^b A(x)\,dx$ or $V = \int_c^d A(y)\,dy$
- Cross-sectional area
  - Disk: $A = \pi(\text{radius})^2$
  - Washer: $A = \pi(\text{outer radius})^2 - \pi(\text{inner radius})^2$





Disk/Washer Method

Definition (Disk Method)

The volume of the region below $f(x)$ rotated about the x-axis is given by

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} \pi[f(x_i^*)]^2 \Delta x = \int_a^b \pi[f(x)]^2\, dx.$$

Definition (Washer Method)

The volume of the region between curves $f(x)$ and $g(x)$ where $|f(x)| \geq |g(x)|$ rotated about the x-axis is given by

$$\begin{aligned} V &= \lim_{n\to\infty} \sum_{i=1}^{n} \pi[f(x_i^*)]^2 - \pi[g(x_i^*)]^2 \Delta x \\ &= \int_a^b \pi[f(x)]^2 - \pi[g(x)]^2\, dx. \end{aligned}$$

Steps to Compute Volume of a Solid of Revolution

1. Draw a picture
   - Find points of intersection if needed
   - Indicate axis of revolution
2. Find cross-sectional area
   - Determine radius/radii of disk/washer
   - Make a guess of the volume of the solid
3. Set up volume as a definite integral
4. Evaluate the integral(s)
   - Check to see if your answer is reasonable

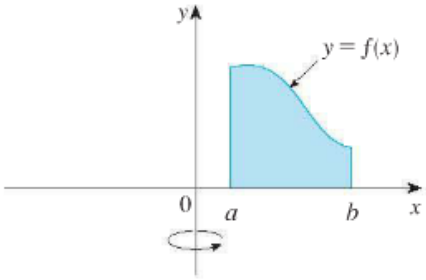
1. Derive the volume formula for the 3D shapes.
   a. $V = \pi r^2 h$
   b. $V = \frac{4}{3}\pi r^3$

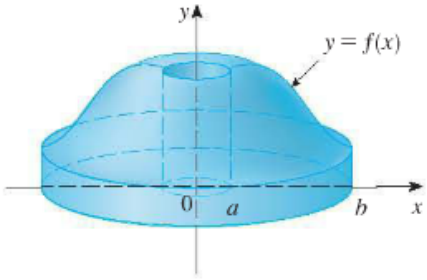
2. Find the volume of the region bounded by $y = \ln x, y = 1, y = 2, x = 0$ about the y-axis.

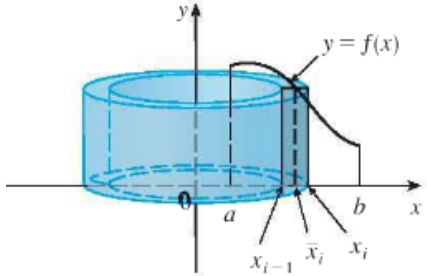
3. Find the volume of the solids obtained by rotating the region bounded by the curves $y = x$ and $y = x^2$ about the following lines.

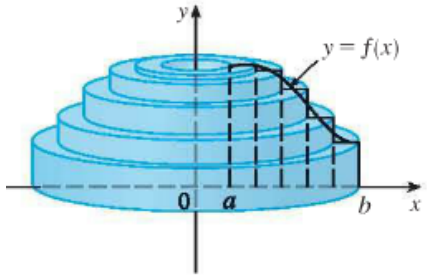
a. x-axis
b. y-axis
c. $y = 2$
d. $x = -1$

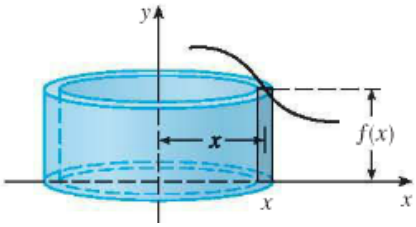
Cylindrical Shells Method

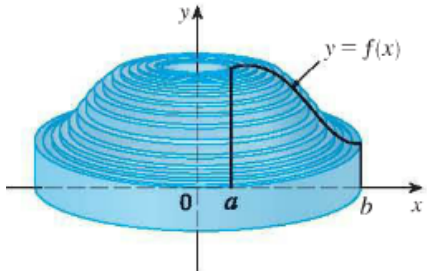










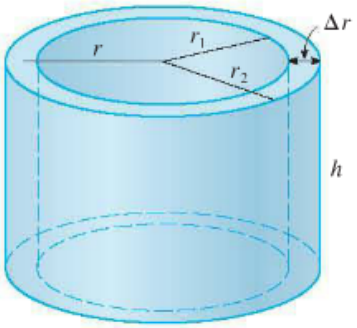


Cylindrical Shells Method

Definition

The volume of the region between $f(x)$ and $g(x)$ (where $f(x) \geq g(x)$) rotated about the y-axis is given by

$$V = \lim_{n\to\infty} \sum_{i=1}^{n} 2\pi x_i [f(x_i^*) - g(x_i^*)]\Delta x$$
$$= \int_a^b 2\pi x [f(x) - g(x)]\, dx.$$



4. Find the volume obtained by rotating the region bounded by $y = 2x^2 - x^3$ and $y = 0$ about the y-axis.

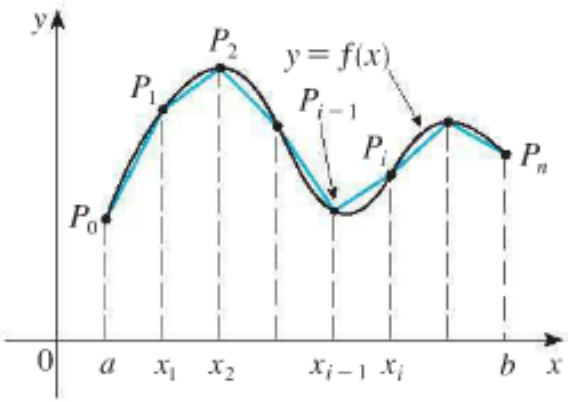
5. Find the volume obtained by rotating the region bounded by $y = e^{-x^2}, y = 0, x = 0, x = 1$ about the y-axis.

6. Find the volume of the solid obtained by rotating the region under the curve $y = \sqrt{x}$ from $x = 0$ to $x = 1$ about the x-axis using

a. disks

b. shells

Arc Length



$$
\begin{aligned}
|P_{i-1}P_i| &= \sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2} \\
&= \sqrt{(\Delta x)^2 + (\Delta y_i)^2} \\
&= \sqrt{(\Delta x)^2 + [f'(x_i^*)\Delta x]^2} \\
&= \sqrt{1 + [f'(x_i^*)]^2}\Delta x
\end{aligned}
$$

Theorem (Arc Length Formula)

If f' is continuous on $[a, b]$, then length of curve $y = f(x)$ is

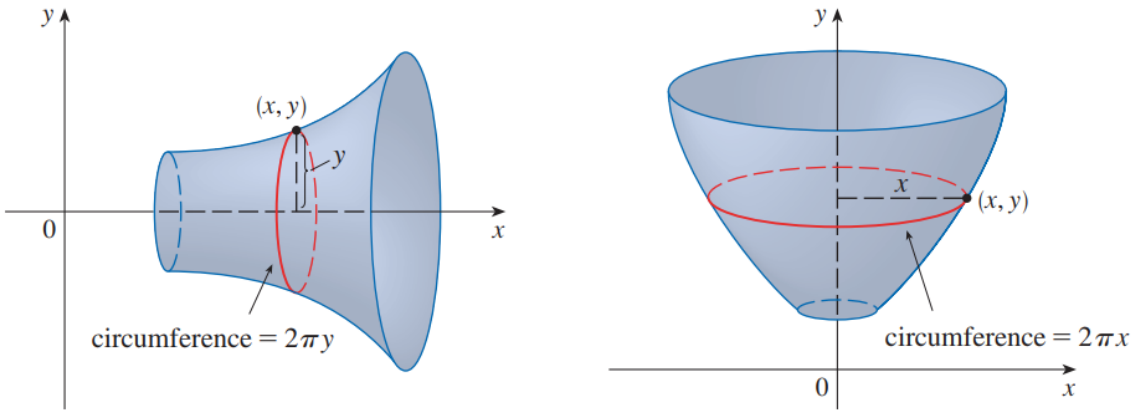
$$L = \lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{1 + [f'(x_i^*)]^2}\Delta x = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx.$$

7. Find the length of the curve

a. $y = \frac{2}{3}(x^2 + 1)^{3/2}$, $[0, 2]$

b. $y = \frac{1}{2}(e^x + e^{-x})$, $[0, 1]$

Area of a Surface of Revolution



(a) Rotation about x-axis: $S = \int 2\pi y \, ds$ (b) Rotation about y-axis: $S = \int 2\pi x \, ds$

Definition

The **surface area** S of the surface obtained by rotating the curve $y = f(x)$ $(a \leq x \leq b)$ about the x-axis is

$$S = \int_a^b 2\pi f(x)\sqrt{1 + [f'(x)]^2}\, dx.$$

8. Set-up a definite integral to represent the surface area obtained by rotating

a. $y = \sqrt{x}$ $(2 \leq x \leq 6)$ about the x-axis

b. $x = 1 + 2y^2$ $(1 \leq y \leq 2)$ about the x-axis

Practice

1. Find the volume of the solid obtained by rotating the region bounded by the curves about the given line:
   a. $y = x^{2/3}, y = 0, x = 1$ about the y-axis
   b. $y = 1/x, y = 0, x = 1, x = 3$ about the $y = -1$
   c. $x = 2\sqrt{y}, x = 0, y = 9$ about the y-axis
   d. $y = e^{-x}, y = 1, x = 2$ about $y = 2$
2. Let R be the region bounded by curves $y = x^2$ and $y^2 = x$. Find the volume obtained by revolving R about the line $y = 4$ by
   a. washers
   b. shells
3. Let R be the region bounded by curves $y = x + 4/x$ and $y = 5$. Find the volume obtained by revolving R about
   a. x-axis
   b. $x = -1$

Practice

4. Set up and simplify integrals that give the length of the given curves. Do not evaluate the integral.
   a. $y = x^2$, $0 \leq x \leq 1$
   b. $y = x^{4/3}$, $-1 \leq x \leq 1$
   c. $y = \ln(\cos x)$, $0 \leq x \leq \frac{\pi}{4}$
5. Find the lengths of curves given.
   a. $y = \frac{1}{6}x^3 + \frac{1}{2x}$ from $x = 1$ to $x = 3$
   b. $y = \frac{1}{2}(e^x + e^{-x})$ from $x = 0$ to $x = 1$
6. Write down an integral which computes the circumference of an ellipse that has semiaxes a and b.
7. Write an integral equal to the surface area of $y = x^2$ from $(1, 1)$ to $(2, 4)$ rotated about the y-axis.
   a. with respect to x
   b. with respect to y

Textbook Exercises (Optional)

6.1: 5, 14, 33, 45, 51
6.2: 1, 3, 13, 28, 37, 42
6.3: 5, 13, 19, 21, 22
6.4: 13, 19, 25, 29

Applications of Integration Summary

- Volumes of revolution can be computed by definite integrals via the disk/washer and/or cylindrical shells methods.
- Many simple curves have arc lengths and surface areas that are impossible to compute in terms of elementary functions.
- What about differential equations?
  TBD in Part II!